



Shoulder Diagnosis and Decision-Making

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Perhaps more than any other joint in the body, the shoulder can present a complex diagnostic challenge to the examining physician. There are a variety of anatomic and clinical reasons for this challenge. First, the “shoulder” is actually a functional complex of four distinct and separate articulations: the sternoclavicular joint, acromioclavicular joint, glenohumeral joint, and scapulothoracic joint. These joints all function together to allow a strong, stable platform with a wide arc of motion through multiple geometric planes. Thus it is not uncommon for patients to present with periscapular pain, for instance, which is actually secondary to a primary glenohumeral problem such as adhesive capsulitis or a large rotator cuff tear.

“Shoulder pain” can also be due to referred pain from the neck. Cervical radiculopathies, paracervical muscle strains, “whiplash,” and other neck-related pathologies can all cause patients to present to the clinician with a subjective feeling of pain in their “shoulder.” There are also vascular etiologies for shoulder pain, peripheral neuropathies and nerve entrapments, intrathoracic pathologies, and malignancies whose presenting symptoms are often shoulder pain or instability.

As a result of the diagnostic challenge presented by patients with shoulder-related symptoms, it is important to take a detailed history, perform a truly regimented and comprehensive physical examination, order and review appropriate diagnostic imaging, and carefully correlate all relevant findings prior to arriving at a final “diagnosis.” In some instances, the selective use of injections can often confirm the etiology of the patient’s pain much more accurately than high-cost advanced imaging techniques. The clinician cannot simply “get an MRI” (magnetic resonance imaging) to arrive at the correct source of the patient’s symptoms, as might be done with other types of musculoskeletal pathologies. In fact, a shoulder MRI may show four or five structural “abnormalities,” none of which have anything to do with the cause of the patient’s presenting symptoms. No part of the diagnostic workup can be taken in isolation. It is only after careful consideration of all the clinical data that a diagnosis and treatment plan can be decided.

The goal of this chapter is to present a practical approach to the patient with shoulder-related symptoms. It will present the diagnostic clues that can be obtained from taking a focused, detailed history, the specific parts of the physical examination that lead to a specific diagnosis, the indications for advanced imaging techniques, the use of selective diagnostic and therapeutic

injections that can confirm the etiology of the patient’s shoulder complaints, and how these decisions can be made.

For most patients with shoulder complaints, it is helpful to divide the possible etiologies and diagnoses into the most likely diagnostic categories. These include (1) referred pain from the cervical spine, (2) acromioclavicular (AC) joint pathology, (3) rotator cuff–related pathology, (4) biceps-related pathology, (5) glenohumeral instability, and (6) glenohumeral arthritis. Although pathologies such as occult tumors, traumatic fractures, vascular disease, and peripheral nerve entrapments can also cause shoulder symptoms, they are not the primary focus of this chapter.

HISTORY

The most important parts of a patient’s history are the patient’s age and whether or not there was a specific trauma or incident involved in the onset of symptoms. For patients younger than 25 years of age, the most likely diagnosis is glenohumeral instability or AC joint–related pain. For patients older than 40 years, rotator cuff–related pathology, with or without biceps tendon involvement, is most common.

The next part of the history should be a precise determination of the patient’s chief complaint. Is the reason for the patient’s visit pain, instability, loss of motion, or weakness? Patients can often tell the clinician their own diagnosis if proper questioning is performed in this manner. Loss of motion indicates adhesive capsulitis, large rotator cuff tears, or glenohumeral arthrosis. Weakness is typically due to rotator cuff tears or, less likely, suprascapular nerve entrapment. Instability symptoms are typically related to glenohumeral labral tears or capsular redundancy. “Pain” is less specific but is most commonly due to subacromial impingement (specifically bursitis, rotator cuff tendinitis), biceps tendinitis, partial- or full-thickness rotator cuff tears, and AC joint arthritis. Glenohumeral instability can also present as pain, which is when the patient’s age should also be considered.

If “pain” is the chief complaint, the question of referred pain must be considered. It is helpful to ask the patient to “put one finger where it hurts the most.” If the patient points to his or her AC joint, the diagnosis is typically made. If specific localization is made to the biceps groove, then biceps pathology with or without rotator cuff involvement is common. Pointing to the lateral shoulder near the insertion of the lateral deltoid is indicative of rotator cuff–related pain. However, if the patient cannot

Abstract

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Keywords

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“localize” the pain, the question of referred pain should be considered. When asked to localize the “shoulder” pain that is actually being referred from a cervical spine etiology, patients cannot use one specific finger to localize the pain. Instead, they use their palm to rub over their trapezius and upper arm. As a general rule, truly shoulder-related pain does not radiate below the elbow. If the patient indicates that symptoms travel below the elbow into the hand, then cervical stenosis or cervical disc herniation should be considered.

After the chief complaint is ascertained, the patient should be questioned about any history of trauma or inciting event. If there is no history of trauma, repetitive overuse injuries, tendinopathies and AC joint or glenohumeral arthrosis are most common. If there is a history of trauma, the question of fracture, labral tear, or rotator cuff tear becomes more important. It is helpful to then ask what position the patient’s arm was in when the trauma occurred. If the patient fell with his or her arm at his or her side, an AC joint or clavicular injury is most common. In contrast, falling on an outstretched arm can lead to rotator cuff tears, superior labral tears, and traumatic glenohumeral instability episodes.

Patients should then be asked what position of their arm brings on their symptoms or worsens their pain. Symptoms that worsen when reaching overhead or at night indicate rotator cuff and/or bursal inflammation. Abduction and external rotation worsens symptoms of anterior instability, whereas pain reaching out to the side is often an indicator of biceps-related pain. If reaching across the body is problematic, AC joint arthritis should be considered.

PHYSICAL EXAMINATION

When assessing a patient with “shoulder pain,” the examination should begin with a focused evaluation of the cervical spine to rule out an etiology of referred pain from the neck. First, the patient should be asked to fully flex, extend, and rotate his or her neck from side to side. Restrictions in motion may indicate the presence of cervical spondylosis. The presence of any tenderness along the cervical midline, paraspinal muscles, and trapezius should be assessed. Next, the “Lhermitte sign”¹ is tested, in which there is a generalized electric shock sensation associated with axial compression of the cervical spine. The “Spurling test”² is the most sensitive examination maneuver, in which radicular pain is exacerbated by extension and lateral bending of the neck toward the side of the lesion, which causes further neuroforaminal compromise. A detailed neurologic examination should also be performed to rule out a focal cervical radiculopathy.

The assessment of the shoulder itself is then performed. The patient’s shirt must be removed to permit a thorough inspection of the entire affected shoulder girdle and to compare it with the opposite side. The evaluation begins by inspecting the sternoclavicular joint, clavicle, and acromioclavicular joint for evidence of prominence, swelling, deformity, or discoloration. Previous surgical incisions should be noted, as well as traumatic skin lesions, bruising, and muscular asymmetry. The pectoralis major muscles should be assessed for any asymmetry in their contours, which could indicate a rupture. The biceps muscle is inspected

for any evidence of a Popeye deformity, indicating a torn long head of biceps tendon. Next, any evidence of atrophy in the deltoid muscle is noted, which could indicate a C5 radiculopathy or axillary neuropathy.

It is important to also inspect the posterior aspect of the shoulder girdle. The presence of any atrophy in the supraspinatus and infraspinatus fossae can be diagnostic. For example, if there is evidence of muscle atrophy in the supraspinatus fossa alone, it may indicate the presence of a chronic tear of the supraspinatus tendon. If there is atrophy in both the supraspinatus and infraspinatus fossae, it could indicate a chronic tear of both the supraspinatus and infraspinatus tendons or entrapment of the suprascapular nerve at the level of the suprascapular notch. Isolated atrophy in the infraspinatus fossa alone can indicate entrapment of the suprascapular nerve at the spinoglenoid notch, often from a paralabral cyst.³

Periscapular pathology is best assessed by asking the patient to actively raise his or her arms overhead while observing from behind. Evidence of scapular dyskinesis should be noted, as should evidence of scapular winging.⁴ If the medial border of the scapula becomes prominent with either forward elevation or doing a “push up against the wall,” the examiner should consider an injury to the spinal accessory nerve. Such an injury typically occurs during cervical lymph node biopsies. If the inferior border becomes prominent, injury to the long thoracic nerve is considered. Scapular winging can also contribute to symptoms of glenohumeral pain and instability. The examiner can confirm the contribution of the scapular dyskinesis to the patient’s presenting symptoms by compressing/stabilizing the scapula against the chest wall and asking the patient to again elevate the arm. If this maneuver diminishes the patient’s symptoms or improves active motion, it confirms the scapula as an etiology in the patient’s diagnosis.

Next, palpation of the shoulder girdle should be performed in a systematic manner. The sternoclavicular joint should be palpated for evidence of tenderness or dislocation, followed by the clavicle and acromioclavicular joint. The biceps groove is then palpated just distal to the coracoid process with the arm in neutral rotation. The presence of tenderness typically indicates pathology of the long head of the biceps tendon or attachment of the subscapularis tendon to the lesser tuberosity. Posterior periscapular tenderness can be associated with muscle spasm (“trigger points”) and occasionally cervical radiculopathy. Medial periscapular and trapezial spasms are not uncommon in patients with large rotator cuff tears, adhesive capsulitis, glenohumeral arthrosis, and instability for whom scapulothoracic motion is often used to achieve forward elevation of the arm. Crepitance at the superomedial border of the scapula during active shoulder elevation is seen with scapulothoracic bursitis.

An assessment of the patient’s active and passive range of motion (ROM) is then performed. Specifically, notation should be made of forward elevation, internal rotation, and external rotation. In throwing athletes, it is important to check external rotation at both 90 degrees of glenohumeral abduction, as well as adduction. The “total arc of motion” should then be compared with the opposite shoulder, noting the presence of deficits in elevation or rotation.

Differences between active and passive motion can also indicate a potential diagnosis. If the patient has full passive motion but limited active motion, the presence of a large rotator cuff tear should be considered. If there is limited active and passive motion with a “firm end point,” adhesive capsulitis is often the etiology. If there is limited active and passive motion in the presence of glenohumeral crepitus or a “gear shifting” sensation, glenohumeral arthrosis is typically present. Excess passive external rotation is seen when there is a rupture of the subscapularis tendon.

Strength testing is the next step in the examination of the shoulder. It is used to investigate the integrity of the rotator cuff or the presence of neurologic pathology. The neurologic motor examination of the upper extremity should be completed during the assessment of the cervical spine. At this stage, the supraspinatus muscle function is isolated by performing the “Jobe test” or “empty can” test.⁵ The shoulder is tested in the scapular plane with the elbow extended and with the shoulder in a position of 90 degrees abduction, 30 degrees flexion, and full internal rotation. The patient is asked to resist downward pressure on the forearm by the examiner. Especially in athletes, it is helpful to assess both arms at the same time to identify often subtle differences in strength. The more posterior rotator cuff muscles are tested by resisted external rotation with the arm adducted and in neutral rotation. Weakness in this position is often present with injury to the infraspinatus, whereas weakness in external rotation with the shoulder abducted and externally rotated indicates injury to the teres minor.

It is important not to forget to examine the subscapularis while examining the rotator cuff and its strength. This is performed by the “lift-off,” “belly-press,” and “bear-hug” tests.⁶ The lift-off test is performed by placing the dorsum of the patient’s hand on his or her back and asking him or her to “lift it off.” The result can be difficult to interpret because patients will often use their triceps to perform this task. It can be more reliable for the examiner to lift the hand/arm away from the patient’s back instead, and ask the patient to hold it in that position. The belly-press test is performed by having patients place their hand on their abdomen, wrists in neutral position, and instructing the patients to keep their elbows forward as they press against their abdomen. Gentle pressure can be applied to the elbows to further assess strength. With this maneuver, the latissimus dorsi can substitute for the subscapularis function. The bear-hug test was developed to help isolate testing of the subscapularis.⁷ In this maneuver, patients are asked to place their hand on their opposite shoulder and resist the examiner as he or she starts to pull the elbow anteriorly. When the elbow is lowered, the upper subscapularis is tested. When the elbow is initially elevated, the lower subscapularis is tested.

Lag signs were initially described by Hertel et al.⁸ They are used to help quantify the degree of rotator cuff tearing or dysfunction. The “external rotation lag sign” is performed by asking patients to maintain their arm in a position of maximal passive external rotation with their arm by their side. Any change in position of the arm after it is released by the examiner is then noted and typically indicates a large tear or atrophy of the infraspinatus. The same test is then performed with the shoulder

in a position of 90 degrees of abduction. This is called “horn-blower’s sign” and is used to assess injury to the teres minor.⁹ The lift-off test is also considered a lag sign when testing the subscapularis.

After strength has been tested, specialty tests are performed to narrow the diagnosis further. It is helpful to group the exam maneuvers by shoulder pathology. First, the subacromial space and rotator cuff are further assessed by the “Neer” and “Hawkins” impingement signs. The Neer test¹⁰ is performed by forward flexing the patient’s arm with the elbow extending and hand pronated. The presence of pain at maximal forward elevation is a sign of subacromial impingement, as seen with bursitis and rotator cuff tendinopathy. Hawkins test¹¹ is sensitive for the presence of subacromial impingement and is tested by the examiner internally rotating the arm after placing it in 90 degrees of glenohumeral and elbow flexion.

Shoulder instability is the next diagnostic category to be considered. The examiner needs to perform multiple specialty tests when evaluating for shoulder instability, given the large spectrum of pathology that can be present in patients with an unstable shoulder. Perhaps the most sensitive and pathognomonic test for anterior shoulder instability is the “apprehension test.” This test is best performed with the patient lying supine. In this position, patients tend to be less tense and relax their shoulder musculature to a greater extent. Once supine, the patient’s arm is abducted to 90 degrees and slowly externally rotated while the examiner also exerts a gentle and steady anteriorly directed pressure on the posterior aspect of the shoulder. A positive test is when the patient indicates a feeling of “apprehension” that his or her shoulder may dislocate if external rotation and anterior pressure continues. It is important to differentiate between an expression of true “apprehension” versus “pain,” because pain can be experienced in this position with other shoulder pathologies besides shoulder instability alone. The apprehension test is one of the most specific tests around the shoulder. It is difficult to assign a diagnosis of anterior instability in the absence of a positive apprehension sign. The “relocation test” can support the diagnosis of anterior instability. In this test, the examiner applies a posteriorly directed “relocation/reduction” pressure to the anterior aspect of the shoulder, which decreases the patient’s apprehension and often allows the arm to be further rotated without issue.

The “posterior apprehension test” is performed with the patient supine or seated. The arm is forward flexed to 90 degrees and maximally adducted while the examiner applies a slight posteriorly directed axial load. The test is positive if the patient describes apprehension that the shoulder will dislocate. However, unlike the anterior apprehension test, the posterior apprehension test is much less sensitive for the presence of posterior instability. Because most patients with posterior instability seldom actually dislocate their shoulders posteriorly, there is typically not a learned “apprehensive” response to this position as is seen in anterior instability. Instead, a posterior “jerk test” is more sensitive to identify the presence of posterior instability. The examiner applies a posteriorly directed axial load to the arm in 90 degrees of flexion and full internal rotation while the arm is steadily adducted. A positive test is noted if the humeral head slides or

“jerks” over the posterior glenoid rim. A jerk can also be noted upon reduction of the subluxated humeral head.

The “load and shift test” is used to assess and describe both anterior and posterior humeral head translation in the glenoid fossa. The examiner should note whether the humeral head can be translated to the anterior glenoid rim or over the glenoid rim with the patient supine and the arm at 90 degrees of abduction. The test should be repeated with the arm at both 45 and 90 degrees of abduction. The examiner should then apply a posterior load and note the degree of posterior translation. It is important that the load and shift test also be performed on the contralateral shoulder, because findings can be subtle and difficult to interpret in isolation.

The “sulcus sign” is performed to assess the degree of inferior instability and the integrity of the superior glenohumeral ligament and rotator interval. The examiner exerts an inferiorly directed pull on the arm with it placed at the patient’s side. The presence and size of a dimple or “sulcus” at the edge of the acromion is then noted. The sulcus should be eliminated when the test is repeated with the arm in a position of adduction and external rotation. A persistent sulcus indicates a greater degree of shoulder laxity, especially in the rotator interval.

The presence and extent of labral tearing is further assessed by examining for injuries to the superior labrum. Although multiple tests have been described, the active compression test (“O’Brien test”) is typically regarded as the most sensitive and specific for a torn superior labrum.¹² In this maneuver, the arm is placed in a position of 90 degrees forward flexion, 10 degrees adduction, maximum internal rotation, and elbow extension. The examiner exerts a downward pressure on the patient’s arm, and the presence of pain and/or weakness is assessed. The test is repeated with the hand and forearm maximally supinated. The test is positive if the patient describes feeling a “deep” pain that is relieved or eliminated when the hand is supinated. If pain persists with the hand fully supinated, it often is a sign of AC joint arthritis. AC joint pain can be confirmed by performing a “cross-body adduction test” in which the arm is then maximally adducted across the body and pain is elicited at the AC joint.

The biceps tendon itself is tested by the “Speed test” and “Yergason test.”¹³ The Speed test is performed when the examiner exerts a downward pressure with the patient’s arm extended in 90 degrees of forward flexion and fully supinated. A painful response is indicative of pathology in the biceps groove. Less commonly, it can be positive in the setting of a superior labral tear. The Yergason test is performed with the patient’s arm by his or her side, the elbow flexed 90 degrees, and the forearm pronated. The patient is then asked to supinate the arm against resistance. Although this test is more sensitive for biceps pathology at the elbow, it can elicit pain in the biceps groove for patients with long head biceps tendon pathology as well.

DECISION-MAKING

Referred Pain From the Neck

The most common source of referred pain to the shoulder is cervical spine pathology. Of the various different anatomic diagnoses, the most commonly encountered are cervical strains,

whiplash, cervical stenosis/spondylosis, cervical disc herniation, and “burners/stingers.”

Cervical Strain

Cervical strains are typically the result of repetitive overuse or mild trauma. The patient can present with “shoulder pain,” which, upon further questioning, localizes to the trapezius and paracervical area posteriorly. The mechanism of injury is generally a forced motion or trauma that causes an eccentric contraction of the neck, such as a sudden twist or turn. Important to the history is a lack of true radicular or arm symptoms. On physical examination, there is tenderness over the paraspinal muscles and trapezius and an absence of positive provocative neurologic tests such as the Lhermitte sign or Spurling test. Diagnostic testing should include plain x-rays if there is a significant trauma, including lateral flexion/extension views. There is often a loss of cervical lordosis in the acute stage from paracervical spasm. MRI is seldom indicated in the absence of true radicular complaints, and symptomatic treatment can progress as tolerated.

Whiplash

A whiplash injury is a subset of cervical strains. The patient typically complains of posterior shoulder and neck pain but no arm pain. There is a history of forced flexion and extension of the cervical spine, typically following a motor vehicle accident. In this situation, a careful neurologic examination should be performed to rule out an acute radiculopathy from a herniated cervical disc or instability. Lateral flexion/extension views of the cervical spine should be obtained to rule out ligamentous instability. If present, advanced imaging including an MRI is indicated.

Cervical Spondylosis/Stenosis

Degenerative changes of the cervical spine can lead to symptoms of shoulder pain. There is seldom a recent history of trauma, although one may exist in the distant past. Unlike cervical strain injuries, patients with spondylosis often complain of both neck/shoulder pain and radicular complaints that are worse with extremes of neck motion. On physical examination, the patient will have pain with cervical extension. There can be a positive Lhermitte sign, often with a generalized electric shock sensation associated with axial compression of the cervical spine. The Spurling test is the most sensitive examination maneuver, in which radicular pain is exacerbated by extension and lateral bending of the neck toward the side of the lesion, which causes further neuroforaminal compromise. A detailed neurologic examination should be performed, looking for dermatomal loss of sensation or motor weakness; and decreased biceps, triceps, or brachioradialis reflexes. Plain x-rays show hypertrophy of facet joints that cause central stenosis or foraminal encroachment. MRI is indicated when localizing neurologic signs are obtained on exam, such as dermatomal loss of sensation and weakness in the setting of radicular complaints.

Cervical Disc Herniation

Acute cervical disc herniations presenting as “shoulder pain” are relatively uncommon but do exist. The patient typically presents

with a history of trauma that caused an acute hyperflexion of the cervical spine, followed by true radicular complaints. Less commonly, the disc herniation can cause a feeling of posterior shoulder pain and muscle spasm. Both the Lhermitte sign and Spurling test are generally positive, and an MRI is indicated for further treatment decision analysis.

Burners/Stingers

A burner is an upper cervical root neurapraxia (reduction or block of conduction across a segment of nerve with conservation of underlying nerve function) most common at C5/C6. In sports such as football, the diagnosis of a burner or stinger often leads to the chief complaint of shoulder pain and weakness. The player's head is forced to one side (lateral bend), and the contralateral shoulder is forced downwards; it is also seen with falls from a height. The player will give a history that "my arm went dead." On examination, there is often weakness of shoulder abduction due to axillary nerve involvement at C5 and, less commonly, biceps or triceps weakness. What differentiates this injury from a disc herniation is that the symptoms typically clear after several seconds to minutes, with a complete return of sensation and strength. Plain x-rays are recommended to rule out fracture or instability. MRI is not usually indicated acutely, unless motor weakness does not improve after 24 hours.

Referred Pain From an Intrathoracic Source

A rare but serious source of "shoulder pain" can be due to an intrathoracic etiology such as a hemothorax or pleural effusion. In this setting, there is irritation of the phrenic nerve, which has contributions from C3, C4, and C5 nerve roots. The C3 dermatome is located on the superior aspect of the shoulder. As a result, these patients can present with acute "shoulder pain."

COMMON SHOULDER PATHOLOGIES

Acromioclavicular Joint Pathology

Patients with shoulder pain due to AC joint pathology present with a typical history and physical examination. The history should include whether or not there was a history of trauma, and if so, how did it occur? The AC joint is typically injured by a direct blow to the lateral aspect of the shoulder. The patient should then be asked to "place one finger where you feel the pain the most." The patient will almost invariably point directly to the AC joint. On physical examination, there is tenderness at the AC joint, and pain can be elicited by cross-body adduction. If there is any question about whether the AC is the cause of the pain, a selective injection to the AC joint can be performed. Plain films are indicated in the presence of a traumatic injury to rule out fracture, AC joint widening, or dislocation. Weighted stress views are not typically needed. The x-rays will also help to classify an AC joint dislocation, which will have a direct effect on management decisions.¹⁴ Rockwood type 1 and 2 injuries are usually managed nonoperatively. Types 4, 5, and 6 are indications for surgery, whereas type 3 management can be operative or nonoperative. If the type of AC dislocation cannot be made on plain x-ray alone, a computed tomography (CT) scan may be considered.

Rotator Cuff Pathology Impingement Syndrome

Impingement syndrome is one of the most common causes of shoulder pain. Anatomically, impingement syndrome includes pain caused by rotator cuff tendinitis, subacromial bursitis, and biceps tendinitis. Acromioclavicular arthritis also can contribute to symptoms of impingement in the presence of hypertrophic bone spurs. When considering the diagnosis of impingement syndrome, the most important question to ask the patient is their age. Primary impingement is seldom seen in patients less than 30 years of age. The patient should be asked if they have pain at night or with overhead activity, the primary complaints of patients with rotator cuff-related pain. There is usually a history of repetitive overhead activity; such as seen with throwers, tennis players, swimmers, painters, or electricians. On physical examination, there is a positive Neer sign and Hawkins sign. The patient may or may not have weakness on the "empty can test."

In this situation, an injection test is helpful to differentiate between simple impingement syndrome and a full-thickness rotator cuff tear. An injection test can be performed using a 25-gauge, 1.5" needle with 9 cc 1% lidocaine and with or without 1 cc steroid. A posterior approach is recommended, approximately 2 cm distal to posterolateral corner of the acromion. If the patient had pain and weakness prior to the injection but no pain or weakness afterwards, then there is not likely to be a torn rotator cuff. Plain x-rays alone are indicated to rule out arthritis or calcific tendinitis. On the other hand, if the patient's pain is gone but he or she still has weakness, then an MRI is indicated to investigate a torn rotator cuff.

Rotator Cuff Tear

Continuing along the spectrum of rotator cuff pathology are rotator cuff tears. The tear can be partial thickness or full thickness, with size and chronicity of the tear playing a major role in the decision-making of the clinician. Classically, the patient has a history of pain at night and with overhead use. There is often a history of trauma but not always. On physical examination, Neer and Hawkins impingement signs are usually positive and there is weakness with supraspinatus ("empty can") or external rotation resistance. There is a discrepancy between active and passive ROM for large and massive tears. In this setting, plain x-rays and an MRI are indicated. The MRI in particular is useful to help differentiate between partial- and full-thickness tears, labral tears, the presence of associated biceps lesions, the size of the tear, and whether any fatty atrophy of the rotator cuff is present. MRI also allows visualization of the subscapularis, which is often overlooked when evaluating patients with shoulder pain and weakness.

Shoulder instability. Shoulder instability is most commonly seen in patients younger than 25 years of age. It can result from both traumatic and atraumatic etiologies. Patients may present with a history of a single instability episode following an injury, or multiple subluxation events from a positional or even voluntary etiology. Shoulder instability occurs along a spectrum of severity. In fact, chief complaints of shoulder "pain" are more

common than “instability.” To help make the diagnosis of shoulder instability, several specific questions are helpful. For example, “Does your shoulder feel loose?” or “Have you ever dislocated your shoulder?” can lead to an immediate diagnosis. In addition, “Do you avoid placing your arm in certain positions?” and “Do you have difficulty reaching behind you, throwing, or pushing open a heavy door?” can also help to determine if anterior or posterior instability is present. “Is it difficult to lift a heavy bag or suitcase?” also indicates whether an inferior or multidirectional component are present.

On physical examination, the apprehension test is pathognomonic for anterior instability. The relocation test, in which the aforementioned apprehensive sensation is eliminated with posteriorly directed pressure on the joint by the examiner, helps to confirm the diagnosis. An anterior and posterior load test is similar to the Lachman test in the knee, in which a manual attempt to translate joint is performed. This test can be difficult to interpret, and a comparison to the contralateral shoulder can be helpful. Lastly, a “sulcus sign” can be performed.

Imaging tests are mandatory when considering the diagnosis of shoulder instability. Plain radiographs are first performed to assess for osseous lesions of the humeral head and glenoid and to confirm that the joint is reduced following a trauma. Specific x-ray views should be considered. First, actual reduction of the glenohumeral joint can only be confirmed by obtaining an axillary view or transscapular Y-view. An anteroposterior (AP) view alone is not acceptable. A Stryker notch view will help to assess for the presence of a Hill-Sachs deformity, whereas a West Point view will identify a bony Bankart lesion. MRI is useful to identify labral tears, capsular tears, or subscapularis tears that are contributing to the instability problem. If bony lesions are present, the physician may consider a CT scan with or without three-dimensional reconstructions to help determine whether an arthroscopic versus open approach to a shoulder stabilization is optimal.

Adhesive capsulitis. Adhesive capsulitis is most commonly seen in diabetic patients. Typically, there is an insidious onset of motion loss following a period of pain or “tendonitis.” Patients have a chief complaint of pain and restricted motion. Early in its course, the pain is often more severe than that seen in patients with rotator cuff tears, arthritis, or simple impingement. The natural history is typically 18 to 24 months to restore motion and decrease pain, if untreated. On physical examination, patients have restricted active and passive motion, as opposed to patients with a torn rotator cuff who have restricted active motion but full passive motion. A true AP x-ray and axillary view should be obtained to rule out osteoarthritis and calcific tendinitis.

Superior Labrum, Anterior to Posterior tears. Superior Labrum, Anterior to Posterior (SLAP) tears typically occur after a history of an eccentric contraction of the biceps muscle. This mechanism is typically caused by a fall on an outstretched arm, or during the deceleration phase of throwing. Patients will present with a history of either anterior or posterior shoulder pain, “rotator cuff symptoms” such as pain at night and with overhead

activity, and classically pain when reaching out to the side. There are multiple tests that have been described to test for a SLAP tear, but the active compression test (O’Brien test) is the most sensitive.¹⁵ MRI with or without contrast helps to confirm the diagnosis, while ruling out confounding pathology such as a torn rotator cuff.

Glenohumeral arthritis. The shoulder is typically not a “weight-bearing joint.” Therefore shoulder arthritis is not as common as hip or knee arthritis. Patients with glenohumeral arthritis typically present with complaints of posterior shoulder pain, limited motion, and difficulty sleeping. On physical examination, patients have glenohumeral crepitus, with limited ROM. The diagnosis is confirmed by plain radiographs. Specifically, a true AP x-ray of the glenohumeral joint (“Grashey view”) is important, and an axillary view helps to confirm the diagnosis and assess for joint subluxation.

“Shoulder pain” can be a complex complaint to evaluate. It should be remembered that shoulder pathologies can be present in tandem, with a multifactorial basis for the patient’s presenting complaints of pain and disability. An accurate diagnosis following injury to the shoulder requires a careful, focused history, a comprehensive physical examination, and judicious use of appropriate imaging modalities.

For a complete list of references, go to ExpertConsult.com.

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Level of Evidence:

IV

Summary:

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Citation:

Hertel R, Ballmer FT, Lombert SM, et al. Lag signs in the diagnosis of rotator cuff rupture. *J Shoulder Elbow Surg*. 1996;5(4):307–313.

Level of Evidence:

V

Summary:

Original description of lag signs in the diagnosis of rotator cuff tears.

Citation:

Lee S, Savin DD, Shah NR, et al. Scapular winging: evaluation and treatment: AAOS exhibit selection. *J Bone Joint Surg Am*. 2015;21(97):1708–1716.

Level of Evidence:

V

Summary:

Reviews the evaluation of different types of scapular winging and their treatment options.

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